

Document

Voice-Based Government Scheme Assistant

1. System Overview

This document describes the architecture, control flow, and memory design of a voice-first government scheme eligibility assistant built using a Planner–Executor–Evaluator agent.

2. Architecture Components

Frontend

A web interface built using **HTML** and **JavaScript** that:

- Records the user's voice
- Sends audio to the backend
- Plays the agent's text-to-speech (TTS) response
- Displays the chat history

Backend (FastAPI)

Responsible for:

- Receiving audio input from the frontend
- Converting speech to text using **Whisper**
- Sending text input to the agent
- Converting the agent's text response to audio using **gTTS**
- Returning text and audio file paths as JSON

Agent (Planner–Executor–Evaluator)

Planner

- Checks for missing user details such as **age**, **income**, and **occupation**
- Asks follow-up questions when required

Executor

- Uses tools to compute the list of eligible schemes

Evaluator

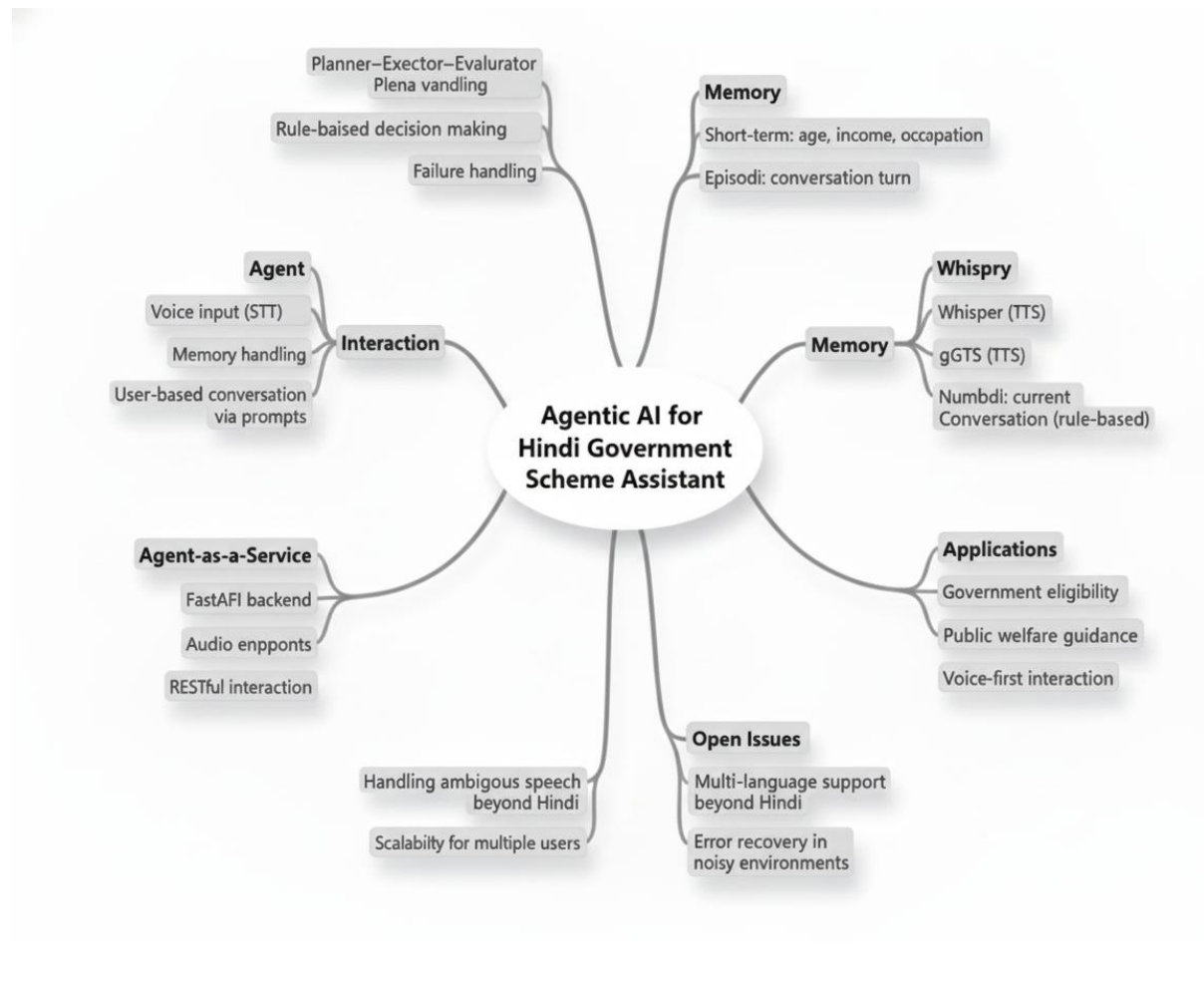
- Formats the final response
- Fetches detailed information about the schemes

Memory

- Stores user information across multiple conversation turns
- Detects contradictions
- Updates or clears stored data when necessary

Tools

- **eligibility_tool**: Determines eligible schemes based on user data
- **scheme_info_tool**: Retrieves detailed information about each scheme



3. Agent Control Flow

1. Input Interpretation

- After Whisper converts voice to text, the agent checks whether the input is usable
- If the text is empty or unclear, execution is stopped and the user is asked to repeat
- This acts as **failure gating**, preventing incorrect reasoning on noisy input

2. Information Extraction

- The agent checks whether it is expecting a specific field (age, income, or occupation)
- If yes, it performs **rule-based extraction** using:
 - Digit parsing
 - Hindi number-word mapping
 - Keyword matching
- This ensures controlled extraction instead of free-text guessing

3. Planner Decision

- The planner inspects conversation memory to identify missing required fields
- If any field is missing, it asks only for the **next required field**
- Otherwise, it proceeds to execution
- This enforces an ordered, deterministic flow similar to a **finite-state controller**

4. Tool Invocation

- Tools are invoked only when all required information is available
- The agent calls `eligibility_tool()` to compute scheme eligibility
- This **tool-gated reasoning** avoids premature or incorrect decisions

5. Evaluation & Response

- The agent evaluates tool outputs
- If no schemes are applicable, an ineligibility response is returned
- Otherwise, scheme details are fetched and explained in a clear, user-safe manner

6. Output Modality

- The final response is converted to **Hindi speech** using TTS
- Both text and audio are returned to the frontend
- Ensures a **voice-first interaction**

4. Memory Design

Stored Fields

The memory stores only four fields:

- **age** → User's age (number)
- **income** → User's yearly income (number)
- **occupation** → User's job category (text)
- **expected_field** → The field the agent is waiting for next

This minimal design keeps the system **simple, stable, and explainable**.

How Memory Is Used During Conversation

Each time the user speaks:

1. The agent reads existing memory
2. Attempts to extract new information from the current input
3. Updates memory if valid data is found
4. Decides what to ask next or whether eligibility can be computed

This ensures the conversation progresses **step-by-step**, not randomly.

Key Properties

- Prevents random questioning
- Enforces a strict order
- Guarantees convergence

This behavior is equivalent to a **finite-state controller**, not a generic chatbot.

5. Prompt Categories

I. Information-Request Prompts

Used by the planner when required data is missing.

Examples:

- कृपया अपनी उम्र बताएं।
 - कृपया अपनी income बताएं।
 - कृपया अपनी occupation बताएं।
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II. Failure-Recovery Prompts

Used when speech input is unclear or cannot be processed.

Example:

- माफ़ कीजिए, आपकी आवाज़ समझ नहीं आई। कृपया दोबारा बोलें।
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III. Eligibility Result Prompts

Used by the evaluator after tool execution.

Examples:

- आप इन योजनाओं के लिए पात्र हैं।

- आप किसी भी सरकारी योजना के लिए पात्र नहीं हैं।

IV. Scheme Explanation Prompts

Used to explain selected schemes to the user.

Example:

- आयुष्मान भारत योजना मुफ्त स्वास्थ्य बीमा देती है।

Flow Chart

